

1. **Experiment:** SPR measurement of cytokine-receptor binding affinity
2. **Time:** 2026.06.05-2026.06.29
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4. **Materials:** PBS buffer (pH 7.4) with 0.05% Tween 20 as running buffer, EDC (N-ethyl-N'-(3-dimethylaminopropyl)carbodiimide), NHS (N-hydroxysuccinimide), 1 M ethanolamine (pH 8.5), 10 mM glycine-HCl (pH 2.0) as regeneration solution, anti-His antibody, IL-6/sIL-6R α complex, IL-6, sIL-6R α , designed protein variants (analytes), BSA (for non-specific binding control), Ni-NTA purified protein samples from the first-round screening
5. **Method:**
 - (1) Experimental parameter setup: Analytes were prepared as a concentration series spanning 0.1-100 nM, with 200 μ L of each sample loaded into a 96-well plate. A 2-fold serial dilution was performed starting from 100 nM, yielding eight concentration points: 100, 50, 25, 12.5, 6.25, 3.125, 1.5625, and 0.78125 nM. For each concentration, 200 μ L of sample was prepared and loaded into a 96well plate. The flow rate was set to 30 μ L/min. The association time was set to 180 s, and the dissociation time was set to 300 s; during the dissociation phase, the instrument automatically switched to running buffer flowing continuously over the chip surface. Regeneration was performed using 10 mM glycineHCl (pH 2.0) with a regeneration time of 30 s. After each cycle, regeneration was executed, and the baseline was allowed to return to the initial level before the next sample injection. The experimental temperature was maintained at 25 $^{\circ}$ C.
 - (2) Chip preparation and system equilibration: We installed a new sensor chip into the SPR instrument following the manufacturer's instructions. A total of 1 L of running buffer (PBS + 0.05% Tween 20) was prepared, degassed by sonication for 10 min to remove bubbles, and placed in the buffer reservoir of the instrument. The instrument was started, and the flow rate was set to 30 μ L/min. The running buffer was allowed to flow continuously over the chip surface for approximately 30 min until the baseline signal stabilized at around 10,000 RU, with fluctuations within ± 1 RU, indicating that the system had reached equilibrium.
 - (3) Methods Edit: In the instrument control software, I created a new kinetic experiment and configured the following parameters in order:
 - ① Chip and flow cell definition: Two flow cells were used in the experiment. Flow Cell 1 was designated as the reference cell, and Flow Cell 2 was designated as the active cell for ligand immobilization.
 - ② Anti-His antibody immobilization and candidate-protein capture: A 1:1 mixture of EDC and NHS was injected at a flow rate of 10 μ L/min for 7 min to activate the carboxymethylated dextran surface, followed by injection of the Anti-His antibody diluted to 20 μ g/mL in 10 mM sodium acetate (pH 5.0) at 10 μ L/min until an immobilization level of 5,000 RU was reached, and finally 1 M ethanolamine (pH 8.5) was injected at 10 μ L/min for 7 min to block the remaining reactive groups on the surface.
 - ③ Kinetic data acquisition: The IL-6/sIL-6R α complex, IL-6, sIL-6R α , and BSA were separately injected over the captured candidate protein from low to high concentrations within the 0.1-100 nM range. Following the parameters set in Section A, we programmed the following steps in the software: running buffer was first flowed for 180 s for baseline stabilization, after which the analyte was injected in order from low to high concentrations (0.78125, 1.5625, 3.125, 6.25, 12.5, 25, 50, and 100 nM), each for 180 s to record the association curve; then the analyte injection was stopped and the flow switched to running buffer over the chip surface for 300 s to record the dissociation curve; after each cycle, regeneration was performed by injecting 10 mM glycine- HCl (pH 2.0) at 30 μ L/min for 30 s, followed by a 60 s wash with running buffer, and the baseline was allowed to return to the initial level (fluctuations within ± 1 RU) before the next concentration was injected, with each concentration run in duplicate.
 - ④ Running the experiment: We confirmed that the chip was properly installed, the buffer volume was

sufficient, the sample plate was correctly placed on the injection rack, and the instrument door was securely closed. The run was then started by clicking "Start". The entire experiment ran for approximately 4 h.

- (4) Result Analysis: After the experiment, the data were exported to the data analysis software. We first subtracted the reference cell sensorgram from the active cell sensorgram to correct for bulk refractive index changes and non-specific binding. A 1:1 binding model was selected for global fitting of the sensorgrams from the eight concentration gradients. The software automatically calculated the association rate constant (k_{on}), dissociation rate constant (k_{off}), and affinity constant (K_D). We compared the fitted curves with the experimental curves; the R^2 value of the fitted curves was 0.995, indicating a good quality of fit. The final K_D value obtained for this sample was 8.5×10^{-8} M.

6. Result:

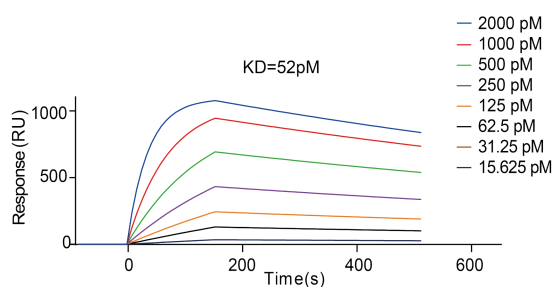


Figure 1. Representative surface plasmon resonance (SPR) sensorgrams showing the binding of STAR6 to the IL-6/sIL-6R α complex at the indicated concentrations (15.625-2,000 pM). Kinetic fitting yielded an equilibrium dissociation constant (K_D) of 52 pM.